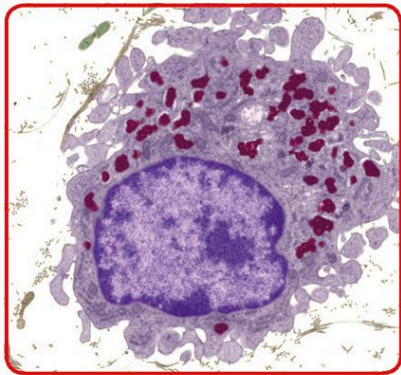


Cells

The living building blocks of animals and plants, cells are the smallest units of life. Even at this microscopic level, each one contains many complex and specialized parts.

Cells need to be complex to perform all the jobs needed for life. They process food, release energy, respond to their surroundings, and—within their minuscule limits—build materials to grow. In different parts of the body, many cells are highly specialized. Cells in the muscles of animals can twitch to move limbs and those in blood are ready to fight infection.



Nucleus

The nucleus (dark purple) controls the activity of the cell. It is packed with DNA (deoxyribonucleic acid)—the cell's genetic material.

Pseudopodium

One of many fingerlike extensions of cytoplasm helps this kind of cell to engulf bacteria.

Cells eating cells

A white blood cell is one of the busiest cells in a human body, part of a miniature army that destroys potentially harmful bacteria. Many white blood cells do this by changing shape to swallow invading cells: they extend fingers of cytoplasm that sweep bacteria into sacs for digestion.

Cytoplasm
The jellylike cytoplasm holds all the cell's parts—known as organelles.

Cell membrane
A thin, oily layer controls the movement of substances into and out of the cell.

Centriole
Structural proteins called microtubules are assembled around a cylindrical arrangement known as a centriole.

Golgi apparatus
The Golgi apparatus packages proteins and sends them to where they are needed.

1 Bacterium approaches

Bacterial cells are 100 times smaller than blood cells, but potentially cause disease. It is a white blood cell's job to prevent them from invading the body.

2 Food vacuole forms

The blood cell envelops bacteria within its cytoplasm, trapping them in fluid-filled sacs called food vacuoles.

3 Digestion begins

Tiny bags of digestive fluid—called lysosomes—fuse with the food vacuole and empty their contents onto the entrapped bacteria.

